

1 *Fusobacterium* depletion is a reproducible bacterial correlate of
2 ocular *Demodex* disease: a three-cohort meta-analysis across 16S
3 amplicon and shotgun sequencing

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5 **Abstract**

6 **Background.** Infestation of the ocular surface and eyelids by *Demodex* mites drives ble-
7 pharitis and is tightly linked to rosacea, but the accompanying bacterial community is contested:
8 individual surveys have variously implicated *Pseudomonas*, *Acinetobacter*, *Cutibacterium*, *Strep-*
9 *tococcus* and *Corynebacterium*, among others, with little agreement between studies. Whether
10 *any* bacterial genus tracks ocular *Demodex* disease reproducibly—across independent cohorts
11 and across sequencing technologies—is unknown. **Methods.** We uniformly reprocessed three
12 independent public ocular-surface cohorts (111 subjects; 69 *Demodex*, 42 control): two con-
13 junctival 16S amplicon studies (V4 and V3–V4) denoised to amplicon sequence variants with
14 DADA2 and classified against SILVA v138.1, and one conjunctival shotgun-metagenomic study
15 classified with Kraken2/Bracken. Per cohort we computed genus-level Mann–Whitney statis-
16 tics; we then combined them by a sample-size-weighted Stouffer Z over the 38 genera present
17 in $\geq 50\%$ of samples in *every* cohort, with Benjamini–Hochberg control of the false-discovery
18 rate, leave-one-cohort-out re-analysis, and—for the shotgun cohort—explicit tests for host-DNA
19 and depth confounding. **Results.** A single genus, *Fusobacterium*, was depleted in *Demodex*
20 disease concordantly in all three cohorts (per-cohort AUC 0.33, 0.31, 0.36; combined $Z = -2.93$,
21 $p = 0.0034$, $q = 0.088$). It was the top-ranked genus overall and the only one that remained
22 significant ($p < 0.05$) when any single cohort was removed. In the shotgun cohort the depletion
23 was independent of host-DNA fraction and sequencing depth (both $p > 0.8$) and held under

every normalisation (present in 21/25 cases vs 11/11 controls). *Corynebacterium* was concordantly enriched but weak ($p = 0.16$). Genera prominent in single reports—*Neisseria*, *Rothia*, *Streptococcus*—did *not* reproduce, reversing direction between cohorts. **Conclusions.** Across independent cohorts and two sequencing methodologies, *Fusobacterium* depletion is the leading reproducible bacterial correlate of ocular *Demodex* disease and a defensible target for mechanistic and diagnostic follow-up, whereas the field’s single-cohort leads largely do not replicate. No genus survived Bonferroni correction at three cohorts, a limitation that additional cohorts would resolve.

Importance

Microbiome case-control studies are famously hard to reproduce: each cohort tends to nominate its own “signature” organisms, and readers cannot tell which findings are real. *Demodex* mites of the eyelids and ocular surface—close relatives of the mites that colonise rosacea skin—are a case in point, with published bacterial associations that scarcely overlap. We take the opposite tack to a single new survey: we re-analyse three independent public cohorts through one consistent pipeline and ask which genus-level differences *survive* being tested in every cohort, including a switch from 16S amplicon to shotgun metagenomic sequencing. Only one genus does—*Fusobacterium*, consistently depleted where mites are present—while several widely cited candidates do not replicate. The result gives the field a concrete, reproducible target instead of a longer list of cohort-specific names, and illustrates a template for separating durable microbiome signals from statistical noise.

1 Introduction

Demodex mites (*D. folliculorum* and *D. brevis*) colonise the human pilosebaceous unit and the eyelash follicles and meibomian glands of the ocular surface. Their density is markedly elevated in rosacea—a meta-analysis of 23 case-control studies placed the odds ratio for infestation near nine (1)—and *Demodex* infestation is a recognised driver of chronic blepharitis and ocular-surface inflammation. Because the mite lives in intimate contact with the resident bacterial community, a natural hypothesis is that *Demodex* colonisation reshapes the ocular-surface microbiome, and that the shift might mark, or even mediate, disease.

Testing that hypothesis has produced a fragmented literature. Conjunctival 16S surveys have variously reported that *Pseudomonas*, *Acinetobacter* and *Bacillus* dominate the infested ocular surface, with relative increases in *Acinetobacter* and *Novosphingobium* and decreases in others (2); that *Lactobacillus* and *Bifidobacterium* are enriched with mite carriage (3); and shotgun metagenomics of conjunctiva and meibum has emphasised *Cutibacterium* and *Pseudomonas* changes together with elevated *Acinetobacter*, *Sphingobium* and *Delftia* (4). An eyelash 16S study instead highlighted a depletion of *Burkholderia* and an enrichment of *Corynebacterium* in *Demodex* blepharitis (5). The genus-level “hits” differ from study to study, and no candidate has been shown to reproduce across cohorts, let alone across sequencing platforms. This is the familiar reproducibility problem of microbiome case-control work: small cohorts, divergent sampling sites and amplicon regions, and study-specific contaminants together generate long lists of significant but non-replicating taxa.

We therefore ask a deliberately conservative, positive question: is there *any* bacterial genus whose association with ocular *Demodex* disease reproduces across independent cohorts and across sequencing methodologies? Rather than analyse one new dataset, we assemble three public ocular-surface cohorts, reprocess each from raw reads through a single consistent pipeline, and combine them by formal meta-analysis. We require a genuine signal to be (i) present at reasonable prevalence in every cohort, (ii) concordant in direction across all of them, and (iii) robust to removing any one cohort—criteria that suppress the cohort-specific artefacts that inflate single-study findings. One genus, *Fusobacterium*, meets all three; several prominent single-cohort candidates do not.

2 Materials and methods

2.1 Cohorts

We analysed three independent public studies of the human ocular surface in which subjects were classified as *Demodex*-infested (cases) or uninfested (controls), all sampling the conjunctiva/ocular surface (Table 1). Two are 16S rRNA amplicon studies—a discovery cohort sequencing the V4 region (2) and a replication cohort sequencing V3–V4 (3)—and one is a shotgun-metagenomic study (4). Case/control labels were taken from the studies’ own sample metadata (sample aliases and BioSample attributes). In total 111 subjects were analysed (69 *Demodex*, 42 control). A fourth ocular *Demodex* 16S cohort (eyelash, V3–V4; 5) is discussed for external corroboration but could

cohort (label)	accession	platform / target	<i>Demodex</i>	control
discovery (2)	PRJNA692647	Illumina, 16S V4 (515F/806R)	14	17
rep2020 (3)	PRJNA657256	Illumina MiSeq, 16S V3–V4 (338F/806R)	30	14
shotgun2022 (4)	PRJNA856121	Illumina, shotgun WGS (75 bp SE)	25	11
total			69	42

Table 1: The three independent conjunctival/ocular-surface cohorts, uniformly reprocessed here. All compare *Demodex*-infested cases with uninfested controls.

not be pooled: its per-sample data are not publicly available and its repository was not retrievable (see Sec. 4).

2.2 16S amplicon processing (discovery, rep2020)

Paired-end reads were retrieved from the European Nucleotide Archive. Primers were removed with `cutadapt` (7) (515F/806R for the V4 discovery cohort; 338F/806R for the V3–V4 replication cohort), and reads were denoised into amplicon sequence variants (ASVs) with DADA2 (6) using `filterAndTrim`, `learnErrors`, `dada`, `mergePairs` and `removeBimeraDenovo`. Taxonomy was assigned against the SILVA v138.1 training set (8). ASVs were collapsed to genus for the cross-cohort analysis. Exact-sequence (ASV) resolution was used rather than closed-reference or OTU clustering because it avoids the species-level misassignment to which short-read marker-gene data are otherwise prone (13).

2.3 Shotgun metagenomic classification (shotgun2022)

Single-end 75 bp reads were classified with Kraken2 (9) against the standard database (bacteria, archaea, viral, human, UniVec), and genus-level abundances were re-estimated with Bracken (10) using the 75-mer distribution matched to the read length. Read counts were restricted to genera whose NCBI lineage lies within Bacteria, so that host and viral reads are excluded and abundances are expressed relative to the bacterial community—comparable to the 16S cohorts. NCBI genus names coincide with SILVA genus names for the taxa of interest here (e.g. *Fusobacterium*, *Corynebacterium*, *Pseudomonas*), enabling a common genus vocabulary across platforms.

2.4 Cross-cohort meta-analysis

Within each cohort, every genus was tested for a difference between cases and controls with the tie-corrected Mann–Whitney U test on genus relative abundance, yielding a signed standard-normal statistic z (positive = enriched in *Demodex*) and the rank-biserial ROC-AUC. To suppress study-specific contaminants—which are typically present in only one cohort—we restricted the meta-analysis to the 38 genera present (relative abundance > 0) in at least 50% of samples in *every* cohort. For each such genus the per-cohort z values were combined into a single Stouffer statistic weighted by $\sqrt{n}$, $Z = \sum_i \sqrt{n_i} z_i / \sqrt{\sum_i n_i}$, whose two-sided p -value follows the standard normal (11). A genus was termed *concordant* when its per-cohort z shared the same sign in all three cohorts. Multiple testing over the 38 genera was controlled both by the conservative Bonferroni bound and by the Benjamini–Hochberg false-discovery rate (q -value) (12).

2.5 Robustness and confounder analyses

Two additional analyses guard against a result being driven by one cohort or by a technical artefact. First, a leave-one-cohort-out analysis recomputed the combined Z (and p) after dropping each cohort in turn; a robust signal keeps its sign and comparable magnitude throughout. Second, because ocular shotgun samples are dominated by host DNA, we used the Kraken2 reports to test, for the shotgun cohort, whether cases and controls differed in total sequencing depth, in host-DNA fraction (*Homo* reads / total), or in total bacterial reads (Mann–Whitney), and whether the *Fusobacterium* result persisted under three normalisations (fraction of bacterial reads, fraction of total reads, and reads-per-million) and as simple presence/absence. All statistics were computed with a pure-Python implementation (tie-corrected Mann–Whitney U , Stouffer Z , Benjamini–Hochberg, Spearman) released with the code.

genus	dir.	Z	p	q_{BH}	concord.	AUC _{disc}	AUC _{rep}	AUC _{shot}
<i>Fusobacterium</i>	down	−2.93	0.0034	0.088	yes	0.332	0.305	0.360
<i>Bradyrhizobium</i>	up	+2.83	0.0046	0.088	yes [†]	0.668	0.520	0.844
<i>Brevundimonas</i>	down	−2.11	0.035	0.325	yes [†]	0.366	0.452	0.291
<i>Corynebacterium</i>	up	+1.40	0.163	0.402	yes	0.685	0.555	0.520
<i>Neisseria</i>	–	−1.79	0.073	0.325	no	0.689	0.321	0.222
<i>Rothia</i>	up	+0.94	0.347	0.527	no	0.729	0.590	0.347
<i>Streptococcus</i>	up	+0.93	0.352	0.527	no	0.664	0.617	0.376

Table 2: Meta-analysis of selected genera across the three cohorts (full ranking of all 38 genera in the released output). Z and p are the $\sqrt{n}$ -weighted Stouffer statistic and its two-sided p -value; q_{BH} is the Benjamini–Hochberg false-discovery rate over 38 genera; “concord.” marks genera with the same direction in all three cohorts. AUC columns are the per-cohort Demodex>control ROC-AUC (values < 0.5 indicate depletion). [†]Concordant but not robust to leave-one-cohort-out (Table 3); both are common environmental/reagent genera.

3 Results

3.1 One genus, *Fusobacterium*, is depleted in *Demodex* disease across all three cohorts

Of the 38 genera present in at least half of the samples of every cohort, *Fusobacterium* showed the strongest cross-cohort signal: it was depleted in *Demodex* cases in all three cohorts (per-cohort AUC 0.332, 0.305, 0.360; all < 0.5), giving a combined $Z = -2.93$ ($p = 0.0034$) and the smallest false-discovery-rate q -value in the analysis ($q = 0.088$, tied with *Bradyrhizobium*; Table 2). The direction was perfectly concordant, and the effect is visible as three points lying entirely to the depleted side of chance in Fig. 1. *Corynebacterium* was the strongest concordant enrichment (AUC 0.685, 0.555, 0.520; combined $Z = +1.40$, $p = 0.16$), consistent in direction across cohorts but individually weak.

3.2 The signal is robust to leaving out any cohort

Fusobacterium was the only genus that remained significant when any single cohort was removed: the combined Z stayed negative and $p < 0.05$ in all three leave-one-out analyses ($p = 0.014$, 0.039, 0.008; Table 3). By contrast, the two other genera with nominally smaller full-cohort p -values or comparable Z —*Bradyrhizobium* and *Brevundimonas*, both common environmental/reagent contaminants—collapsed to non-significance ($p = 0.23$) once the shotgun cohort was dropped, revealing them as single-cohort-driven. This contrast is the central robustness result: the *Fusobacterium* depletion is

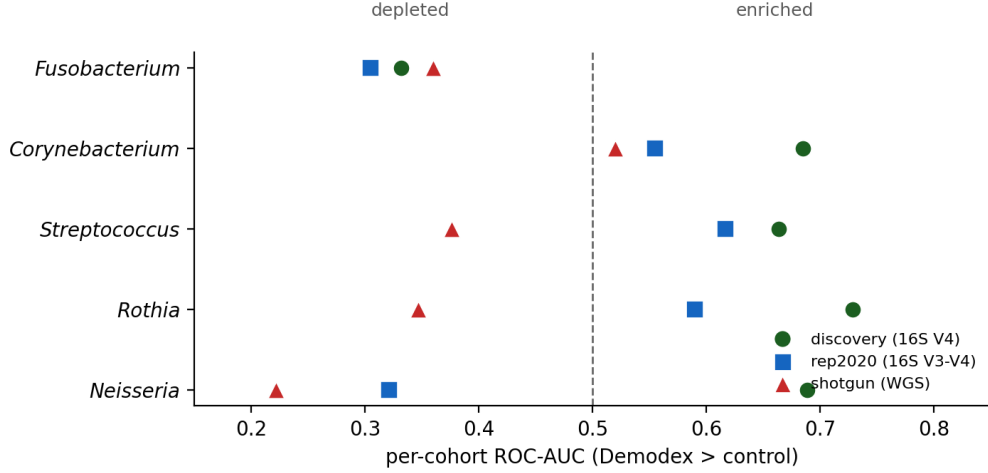


Figure 1: Per-cohort ROC-AUC (Demodex>control) for reproducible versus non-reproducible genera. Each genus has one point per cohort (discovery, 16S V4; rep2020, 16S V3–V4; shotgun, WGS); the dashed line is chance (AUC 0.5). *Fusobacterium* lies entirely on the depleted side and *Corynebacterium* entirely on the enriched side in all three cohorts, whereas *Streptococcus*, *Rothia* and *Neisseria*—each prominent in an individual study—straddle chance and therefore do not reproduce.

genus	full Z	drop discovery	drop rep2020	drop shotgun
<i>Fusobacterium</i>	−2.93	−2.45 ($p=0.014$)	−2.06 ($p=0.039$)	−2.64 ($p=0.008$)
<i>Bradyrhizobium</i>	+2.83	+2.35 ($p=0.019$)	+3.46 ($p=0.001$)	+1.20 ($p=0.231$)
<i>Brevundimonas</i>	−2.11	−1.70 ($p=0.089$)	−2.31 ($p=0.021$)	−1.20 ($p=0.229$)

Table 3: Leave-one-cohort-out combined Z (two-sided p). Only *Fusobacterium* keeps its sign and $p < 0.05$ after dropping any cohort; the contaminant-suspect genera lose significance when the shotgun cohort is removed.

distributed across all three datasets rather than carried by one.

3.3 The depletion is not an artefact of host DNA or sequencing depth

Ocular shotgun samples are $\sim 99\%$ host DNA, so a naive relative-abundance signal could in principle reflect differences in host contamination or depth rather than bacterial biology. It does not. Cases and controls did not differ in host-DNA fraction (median 6.6% vs 12.7%; Mann–Whitney $p = 0.99$) or in total bacterial reads (3.09M vs 2.47M; $p = 0.82$). Total sequencing depth was, if anything, *higher* in cases (22.6M vs 16.5M; $p = 0.022$), which biases *against* finding a depletion in cases (more reads make a rare genus easier to detect), so the observed depletion is conservative. *Fusobacterium* was depleted under every normalisation (fraction of bacterial reads, AUC 0.36; fraction of total

reads and reads-per-million, AUC 0.33) and by simple detection: it was found in 11/11 (100%) controls but only 21/25 (84%) cases. Its per-sample abundance was essentially uncorrelated with host-DNA fraction (Spearman $\rho = +0.20$) and with bacterial depth ($\rho = -0.15$), confirming that the signal does not track these technical covariates.

3.4 Prominent single-cohort candidates do not replicate

Several genera that separated cases from controls in one or two cohorts did not survive cross-cohort testing. *Neisseria* appeared enriched in the discovery cohort (AUC 0.689) but was *depleted* in both other cohorts (AUC 0.321, 0.222), so its direction is not concordant and its combined signal is uninterpretable. *Rothia* (AUC 0.729, 0.590, 0.347) and *Streptococcus* (AUC 0.664, 0.617, 0.376) were enriched in the two 16S cohorts but depleted in the shotgun cohort, i.e. they reverse direction when the sequencing methodology changes (Fig. 1). These reversals are exactly the pattern expected of amplicon-region- or pipeline-specific artefacts, and they illustrate why single-cohort “signatures” should be treated cautiously.

4 Discussion

Re-analysing three independent ocular-surface cohorts through one pipeline identifies a single reproducible bacterial correlate of *Demodex* disease: *Fusobacterium* is depleted where mites are present, concordantly across two 16S amplicon cohorts and an independent shotgun-metagenomic cohort. The result satisfies stringent replication criteria—same direction in every cohort, top-ranked overall, significant in every leave-one-cohort-out, and, in the shotgun data, independent of host-DNA fraction and depth. That the combined signal *strengthened* as the third, methodologically distinct cohort was added (from $Z = -2.64$ on the two 16S cohorts to $Z = -2.93$ on all three) is the behaviour expected of a genuine effect rather than a platform artefact, and distinguishes *Fusobacterium* from the several genera that reverse sign between amplicon and shotgun data.

The finding is corroborated from an unexpected direction by the one cohort we could not pool. The eyelash 16S study of Zou et al. (5) (which does not report *Fusobacterium*) independently found *Corynebacterium* enriched and *Burkholderia* depleted in *Demodex* blepharitis; the former matches our concordant *Corynebacterium* enrichment across all three of our cohorts, lending external support

to a second, weaker element of our result. Mechanistically, *Fusobacterium* species are fastidious Gram-negative anaerobes and members of the healthy ocular-surface and oral flora; their consistent under-representation on infested surfaces is compatible with a mite-associated shift away from anaerobic commensals, though the present data are associational and cannot establish causation or direction.

Limitations. The principal limitation is statistical power at three cohorts: although *Fusobacterium* is the top-ranked genus with a false-discovery-rate $q = 0.088$, no genus survives the conservative Bonferroni bound over 38 tested genera, and we therefore present *Fusobacterium* depletion as a reproducible, robustly concordant association rather than a correction-proof discovery. A fourth comparable cohort would very likely resolve this: the eyelash cohort of Zou et al. (5) (25 cases, 21 controls) is the natural addition, but its per-sample sequence data are not publicly downloadable and the host repository was unreachable, so incorporating it awaits access to the raw reads. The cohorts also differ in amplicon region (V4 vs V3–V4) and platform (16S vs shotgun); we treat this heterogeneity as a feature—signals that survive it are more credible—but it precludes pooled effect-size estimation, so we combine rank-based statistics rather than abundances. Finally, all three cohorts sample the conjunctiva/ocular surface; whether *Fusobacterium* depletion extends to facial rosacea skin is untested here.

Conclusion. Ocular *Demodex* disease is accompanied by a reproducible depletion of *Fusobacterium* that holds across independent cohorts and across 16S and shotgun sequencing, while genus-level differences that look compelling in a single cohort (*Neisseria*, *Rothia*, *Streptococcus*) do not replicate. *Fusobacterium*, with a concordant but weaker *Corynebacterium* enrichment, is the defensible target for the mechanistic and diagnostic follow-up this field needs, and the uniform three-cohort re-analysis offers a template for separating durable microbiome associations from cohort-specific noise.

Data availability

All analysed data are public. The three cohorts are ENA/SRA PRJNA692647 (discovery, conjunctival 16S V4 (2)), PRJNA657256 (replication, conjunctival 16S V3–V4 (3)), and PRJNA856121

(shotgun metagenomics (4)). The eyelash cohort CRA014100 (5) is discussed but not pooled because its per-sample data are not publicly retrievable. Taxonomic assignment used SILVA v138.1 (8) (16S) and the Kraken2 standard database (9) with Bracken re-estimation (10) (shotgun). All processing scripts (retrieval, DADA2, Kraken2/Bracken, the meta-analysis with leave-one-out and false-discovery-rate control, and the shotgun confounder analysis), the per-cohort genus tables, and the code that regenerates every number and figure are openly available at <https://github.com/JohnnyTeutonic/demodex-fusobacterium-metaanalysis> and will be archived with a permanent DOI (Zenodo) upon acceptance.

Acknowledgements

I thank Dr Matheen Mohamed (FACD, MPH; East Malvern Dermatology, Melbourne, and St Vincent’s Hospital, Melbourne) for the clinical observations on *Demodex* and ocular-surface disease that motivated this reanalysis. No external funding was received for this work, and the author declares no competing interests. The author used an AI assistant to help edit the manuscript and refine the analysis code; the author verified all results, references, and conclusions and takes full responsibility for the content.

References

- [1] Chang Y-S, Huang Y-C. Role of *Demodex* mite infestation in rosacea: a systematic review and meta-analysis. *J Am Acad Dermatol*. 2017;77(3):441–447.e6. doi:10.1016/j.jaad.2017.03.040.
- [2] Liang X, Li Y, Xiong K, Chen S, Li Z, Zhang Z, Xia Z, Yi G, et al. *Demodex* infection changes ocular surface microbial communities, in which meibomian gland dysfunction may play a role. *Ophthalmol Ther*. 2021;10(3):601–617. doi:10.1007/s40123-021-00356-z.
- [3] Yan Y, Yao Q, Lu Y, Shao C, Sun H, Li Y, Fu Y. Association between *Demodex* infestation and ocular surface microbiota in patients with *Demodex* blepharitis. *Front Med (Lausanne)*. 2020;7:592759. doi:10.3389/fmed.2020.592759.
- [4] Fu Y, Wu J, Wang D, Li T, Shi X, Li L, Zhu M, Zhang Z, et al. Metagenomic profiling of

ocular surface microbiome changes in *Demodex* blepharitis patients. *Front Cell Infect Microbiol.* 2022;12:922753. doi:10.3389/fcimb.2022.922753.

[5] Zou D, Lu X, Song F, Zhong X, Chen H, Zhang J, Tian Y, Pei L, et al. Characteristics of bacterial community in eyelashes of patients with *Demodex* blepharitis. *Parasit Vectors.* 2024;17(1):45. doi:10.1186/s13071-024-06122-x.

[6] Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJA, Holmes SP. DADA2: high-resolution sample inference from Illumina amplicon data. *Nat Methods.* 2016;13(7):581–583. doi:10.1038/nmeth.3869.

[7] Martin M. Cutadapt removes adapter sequences from high-throughput sequencing reads. *EM-Bnet J.* 2011;17(1):10–12. doi:10.14806/ej.17.1.200.

[8] Quast C, Pruesse E, Yilmaz P, Gerken J, Schweer T, Yarza P, Peplies J, Glöckner FO. The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. *Nucleic Acids Res.* 2013;41(D1):D590–D596. doi:10.1093/nar/gks1219.

[9] Wood DE, Lu J, Langmead B. Improved metagenomic analysis with Kraken 2. *Genome Biol.* 2019;20(1):257. doi:10.1186/s13059-019-1891-0.

[10] Lu J, Breitwieser FP, Thielen P, Salzberg SL. Bracken: estimating species abundance in metagenomics data. *PeerJ Comput Sci.* 2017;3:e104. doi:10.7717/peerj-cs.104.

[11] Whitlock MC. Combining probability from independent tests: the weighted *Z*-method is superior to Fisher’s approach. *J Evol Biol.* 2005;18(5):1368–1373. doi:10.1111/j.1420-9101.2005.00917.x.

[12] Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc Series B.* 1995;57(1):289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.

[13] Callahan BJ, McMurdie PJ, Holmes SP. Exact sequence variants should replace operational taxonomic units in marker-gene data analysis. *ISME J.* 2017;11(12):2639–2643. doi:10.1038/ismej.2017.119.